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Synchronous improvement of methane production and digestate dewaterability in sludge anaerobic digestion by nanobubble

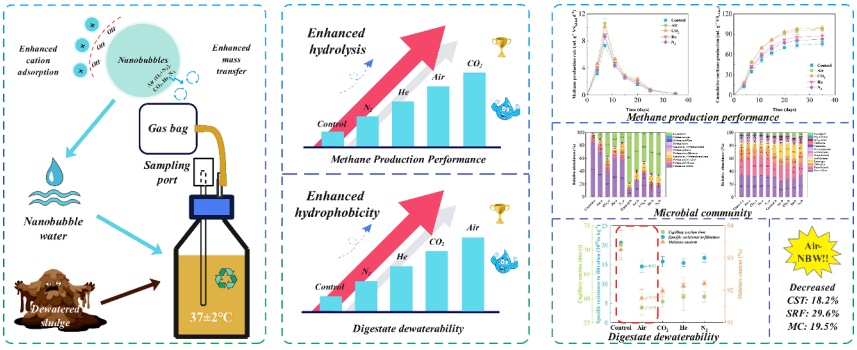
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HIGHLIGHTS

- Four different NBWs were added to the sludge anaerobic digestion.
- The hydrolysis of sludge anaerobic digestion can be enhanced by the addition of NBW.
- Addition of NBW increased methane production by 9.0–27.8% compared with control.
- Addition of NBW (Air, CO_2, He, and N_2) improved the dewaterability of digestate.
- Air-NBW was the most cost-effective additive for sludge anaerobic digestion.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

The subsequence anaerobic digestion (AD) of dewatered sludge (DWS) from wastewater treatment plants necessitates an emphasis on enhancing methane production and dewaterability. The effect of different nanobubble water (NBW) on AD of DWS was investigated under mesophilic conditions. Cumulative methane production was improved by 9.0–27.8% with the addition of different NBW (Air, CO_2, He, and N_2). NBW improved methanogenic performance by significantly enhancing the hydrolysis of sludge AD. Results from the digestate, the capillary suction time, specific resistance to filtration, and moisture content could be decreased by 14.6–18.2%, 18.8–29.6%, and 13.6–19.5%, respectively. The addition of NBW can improve the dewaterability of digestate by reducing the digestate particle size and increasing the zeta potential of digestate. The addition of NBW significantly increased methane production and improved dewaterability in AD; Air-NBW showed the best improvement.

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1. Introduction

Anaerobic digestion (AD) is gaining widespread attention not only for its ability to convert dewatered sludge (DWS) into biogas but also for its ability to reduce environmental pollution significantly (Balasundaram et al., 2023). However, it suffers from poor hydrolysis capacity, easy inhibition, long lag phase, and low biogas production (Mutegoa and Sahini, 2023). To facilitate subsequent processing, digestate usually undergoes dewatering using a belt filter press or a centrifugal dewatering machine (Simoni et al., 2024). Reducing the moisture content (MC: moisture content of DWS cake) of sludge is a key approach to reducing the cost of sludge management (Cao et al., 2023). Consequently, it is essential to discover appropriate techniques to enhance the AD performance and dewaterability of digestate in an economically feasible manner.

The total solid (TS) of the sludge is an important factor during AD. Methane production showed an increasing and then decreasing trend when the TS increased, the optimal TS for AD under mesophilic conditions should be around 8 % (An et al., 2017). However, the TS of DWS is usually up to about 20 % (Simoni et al., 2024). Therefore, the solids content of AD systems can be adjusted by adding other media to the digester. If this media can synchronously improve the sludge hydrolysis, methane yield, and digestate dewaterability, while ensuring the possibility of land use of the digestate, it will have great application prospects. Nanobubble water (NBW) is defined as water containing a large number of bubbles less than 1000 nm in diameter (Kurz et al., 2021), which has some distinct advantages, including robust adsorption capacity, increased gas solubility, extended retention time, and elevated zeta potential in comparison to deionized water (Zinjenab et al., 2023). NBW is non-polluting and its addition to the sludge AD system will not affect the subsequent treatment (Patel et al., 2021).

N_2-NBW increased cumulative methane production by 29.1 % of the sludge AD because the N_2-NBW has a higher electrical charge, which can increase the zeta potential in the system by 17.1 % (Yang et al., 2019). An increase in zeta potential usually implies an increase in repulsive forces between sludge particles, which may result in the sludge flocs becoming more loosely packed, thus facilitating microbial-substrate contact and improving substrate availability, and thus increasing methane production (Marcelino et al., 2023). Air-NBW increased cumulative methane production by 33.0 % of the sludge AD due to the enhanced microbial activity of Air-NBW, which enhances the activity of the electron transport system by 27.2 % (Hou et al., 2022). The benefits of NBW addition may include the promotion of substrate degradation (Fan et al., 2021b), avoiding acid inhibition (Fan et al., 2020), and mitigation of ammonia suppression (Fan et al., 2021b). The positive effects of NBW on methane production from sludge have been revealed in recent research, but its effect on digestate dewaterability is unclear.

After traditional AD, the dewaterability of digestate severely deteriorated, thus improving the dewatering property of digestate has become one of the important tasks in this field (Cao et al., 2023; Simoni et al., 2024). After β-cyclodextrin pretreatment, the hydrolysis rate and cumulative methane production of AD increased by 153.2 % and 175.4 %, respectively; meanwhile, the capillary suction time (CST) of digestate decreased by 24.1 % (Guo et al., 2022). The addition of biochar resulted in improved cumulative methane production (>23.1 %) and decreased digestate specific resistance of filtration (SRF) (>57.1 %) (Liang et al., 2022). The addition of $Ca(OH)_2$ increased cumulative methane production by 33.3 %, improved digestate dewaterability, and decreased moisture content by 7.9 % due to an increase in digestate zeta potential (Li et al., 2017). The high surface charge and theoretically high Laplace pressure of nanobubble induce hydrophobic interaction (Marcelino et al., 2023), which applied to AD may enhance the dewaterability of digestate. The rise in methane generation appeared to be instrumental in enhancing the dewaterability of digestate; however, the impact of NBW addition on the dewaterability of digestate processed anaerobically has not been demonstrated.

Therefore, this study selected four kinds of NBW prepared with cheap and safe gases (Air, CO_2, He, and N_2) to evaluate their effect on methane production of DWS, as well as the influence on the dewaterability of digestate. During the AD process, the methane production and the changes of soluble hydrolysate and acidification products were analyzed. The microbial community subjected to different NBWs was characterized and correlated to the profile of methane production. After digestion, the dewaterability of digestate was analyzed and the role of NBWs addition was discussed.

2. Materials and methods

2.1. Materials

DWS was acquired from a wastewater treatment plant in Lanzhou City, China, using the "oxidation ditch + aerated biofilter process" with a treatment scale of 5000 m^3/d. DWS was cultivated under mesophilic conditions for 10 days to be used as inoculum. The table illustrates the properties of the inoculum and the substrate (see Supplementary Material).

A nanobubble generator (JXWNP-AL01, Jinxiang, China) was utilized to produce different types of NBW. The pressure of the nanobubble generator was maintained at 0.4–0.5 MPa. Air, CO_2, He, and N_2 were introduced into the machine separately to produce NBW using hydrodynamic cavitation. Then the gases were completely mixed with water and allowed to stationary for 10 min at 25 ± 2 °C to form the NBW. The reasons for choosing these four gases are as follows: Air: O_2 can provide micro-oxygenic conditions for AD; CO_2: CO_2 can be involved in the synthesis of hydrogenotrophic methanogenesis; He: He is an inert gas with relatively low molecular weight, and the effect on the AD of sludge is unclear; N_2: N_2 has inert characteristics with relatively high molecular weight (Kurz et al., 2021; Marcelino et al., 2023; Yang et al., 2019).

The AD system was set up as described by Liu et al. (2022). Under mesophilic conditions, DWS was used as the substrate, seed sludge was used as inoculum; deionized water and different NBW were mixed proportionally (the mass mixing ratio was DWS: NBW: seed sludge = 682 g:682 g:136 g), and all of the reactants were loaded into the digesters and gently stirred for 1 min to mix well. The digesters were prepared and put into a water bath (37 ± 2 °C) at the same time, the AD was carried out for a total of 35 days at mesophilic temperature. In this experiment, five experimental groups were set up with the addition of deionized water, Air-NBW, CO_2-NBW, He-NBW, and N_2-NBW. The five conditions were labeled as control, Air, CO_2, He, and N_2 groups, and three parallel experiments were conducted for each condition, with a total of 15 digesters.

2.2. Methods of analysis

On Day 4, 8, 15, 22, 28, and 35, an 80 g sample of the mixture was taken. The samples were centrifuged at 2000 *g* for 5 min, and the supernatant was collected using a 0.45 μm polyethersulfone microfiber filter. The filtered liquid samples were used to determine the indicators.

TS, volatile solid (VS), pH, soluble chemical oxygen demand (SCOD), and total ammonia nitrogen (TAN) were determined by standard methods (APAH,1998), and standard formulas in the literature calculated free ammonia nitrogen (FAN) (Hansen et al., 1998). Volatile fatty acid (VFA) and methane content were determined using a gas chromatograph (SP-7890Plus, Rui Hong, China), and the volume of biogas was determined using an acid gas flow meter (LML-2, Alpha, China).

MC and SRF were measured as described by Zhang et al. (2024). CST was determined using a capillary water absorption meter (304 M, Triton Electronics, UK). The size of the digestate particles and their zeta potential were analyzed using a nanoparticle size and zeta potential analyzer (Nano ZS90, Malvern, UK). The concentration of proteins and polysaccharides in EPS was assayed via a modified Lowry and Anthrone protocol (Chuenchart et al., 2021). Three-dimensional fluorescence

excitation and emission matrix (3D-EEM) scans were obtained using a fluorophotometer (F4600, Hitachi, Japan).

DNA was extracted from digestate, and the microorganisms contained therein were tested for abundance and type using high-throughput sequencing to determine differences in the microbial community in the digesters after adding different NBW. Primers 341F (5′-CCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGGTWTCTAAT-3′) were utilized to amplify the bacterial 16S rRNA gene in the V3-V4 portion of the bacterial 16S rRNA gene; using primers Arch519F (5′-CAGCCGCCGCGGTAA-3′) and Arch915R (5′-GTGCTCCCCCGCCAATTCCT-3′) to amplify the V4 portion of the Archaea V4-V5 portion of the 16S rRNA gene. Each sample was subjected to DNA extraction, followed by sequencing. Finally, initial data and detailed information on species were uploaded through the website (https://www. ncbi.nlm.nih.gov). The accession numbers were PRJNA1066472 and PRJNA1066485.

2.3. Statistical analysis

The data were analyzed using SPSS (27.0), and the data from different experimental groups were analyzed for differences using ANOVA. The $p < 0.05$ indicates a significant difference between the two experimental groups. SPSS was also used to complete the principal component analysis (PCA) and Duncan's test. Pearson correlation analysis was performed for microbial community, dewaterability, and environmental factors. Methane production was fitted to a modified Gompertz model (Balasundaram et al., 2023).

3. Results and discussion

3.1. Anaerobic digestion performance

VFA is an outcome of hydrolysis and acidification and a feedstock for biogas generation (Fan et al., 2021b). The maximum concentration of VFA was found from Day 4 to Day 8, which declined rapidly after Day 8, and stabilized after Day 22 (Fig. 1). On Day 4, the digesters with added NBW (Air, CO_2, and He) significantly increased VFA by 17.9 % ($p < 0.01$), 20.7 % ($p < 0.01$), and 11.2 % ($p < 0.05$) compared with the control. The peak of VFA appeared earlier with the introduction of Air-NBW and CO_2-NBW than with the addition of He-NBW and N_2-NBW, likely as a result of the improved mass transfer performance of Air-NBW and CO_2-NBW (He et al., 2022; Wang et al., 2023). The VFA of CO_2-NBW group was at a high level for the longest period among the five groups. This phenomenon could be attributed to the introduction of CO_2 nanobubbles which enhanced the activity of hydrolyzing bacteria, enhanced the degradation of organic matter, and participated in the process of methane production by the hydrogenotrophic methanogenesis (Chuenchart et al., 2021; Marcelino et al., 2023).

The pH serves as a critical parameter in AD, with a range of 6.5 to 8.0 being optimal for methanogens; an increase in pH will also lead to an increase in FAN (Goodarzi et al., 2024). All digesters showed similar pH trends, with the pH of the digesters ranging from 7.2 to 7.8 (Fig. 2a). As the AD proceeded, the pH value showed a decreasing and then increasing trend, and gradually stabilized, which was caused by the change of VFA in AD (Fan et al., 2021a).

SCOD level indicates the hydrolysis and decomposition of sludge flocs (Liu et al., 2022). All digesters showed the same trend of SCOD

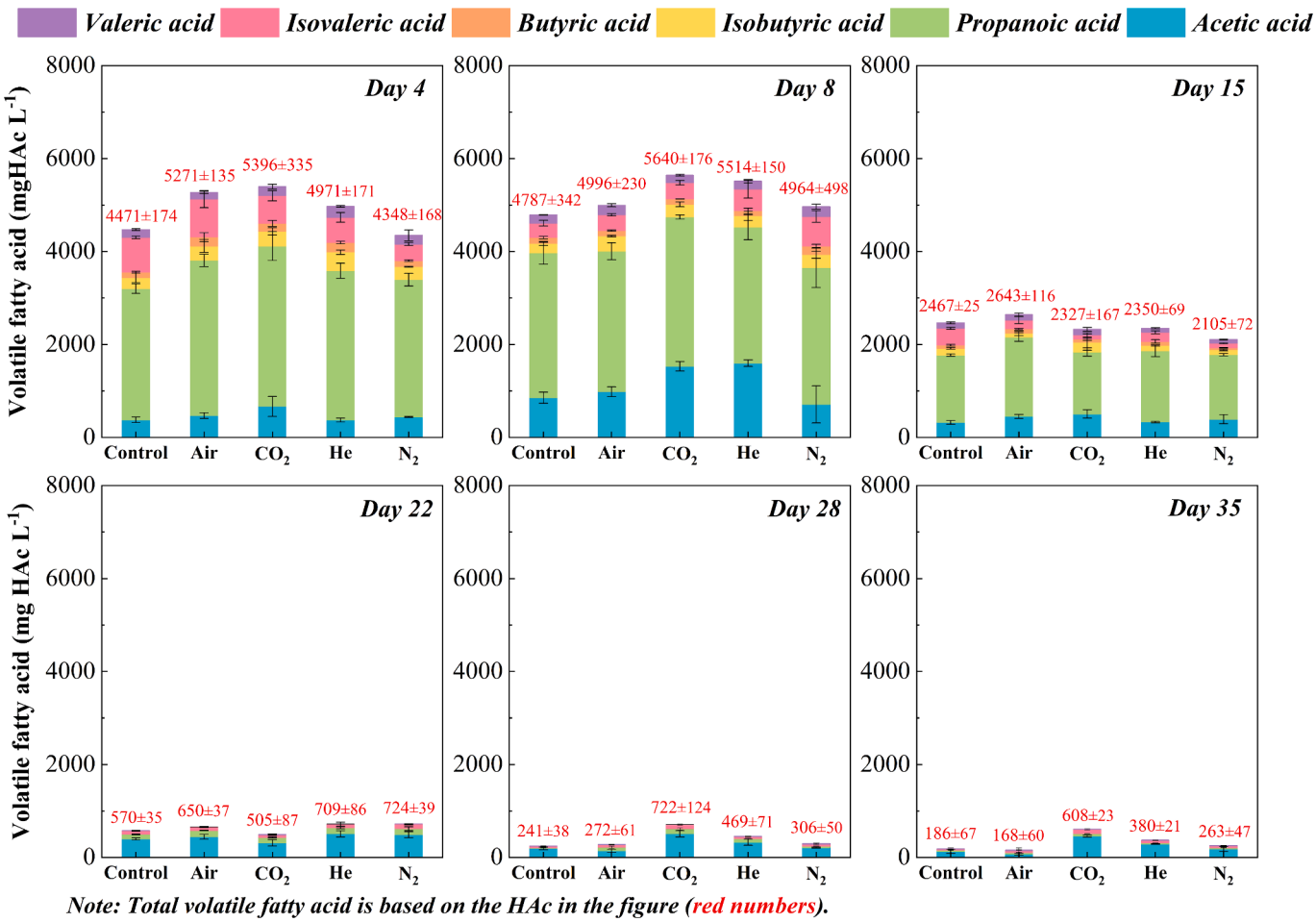


Fig. 1. Changes in VFA during anaerobic digestion.

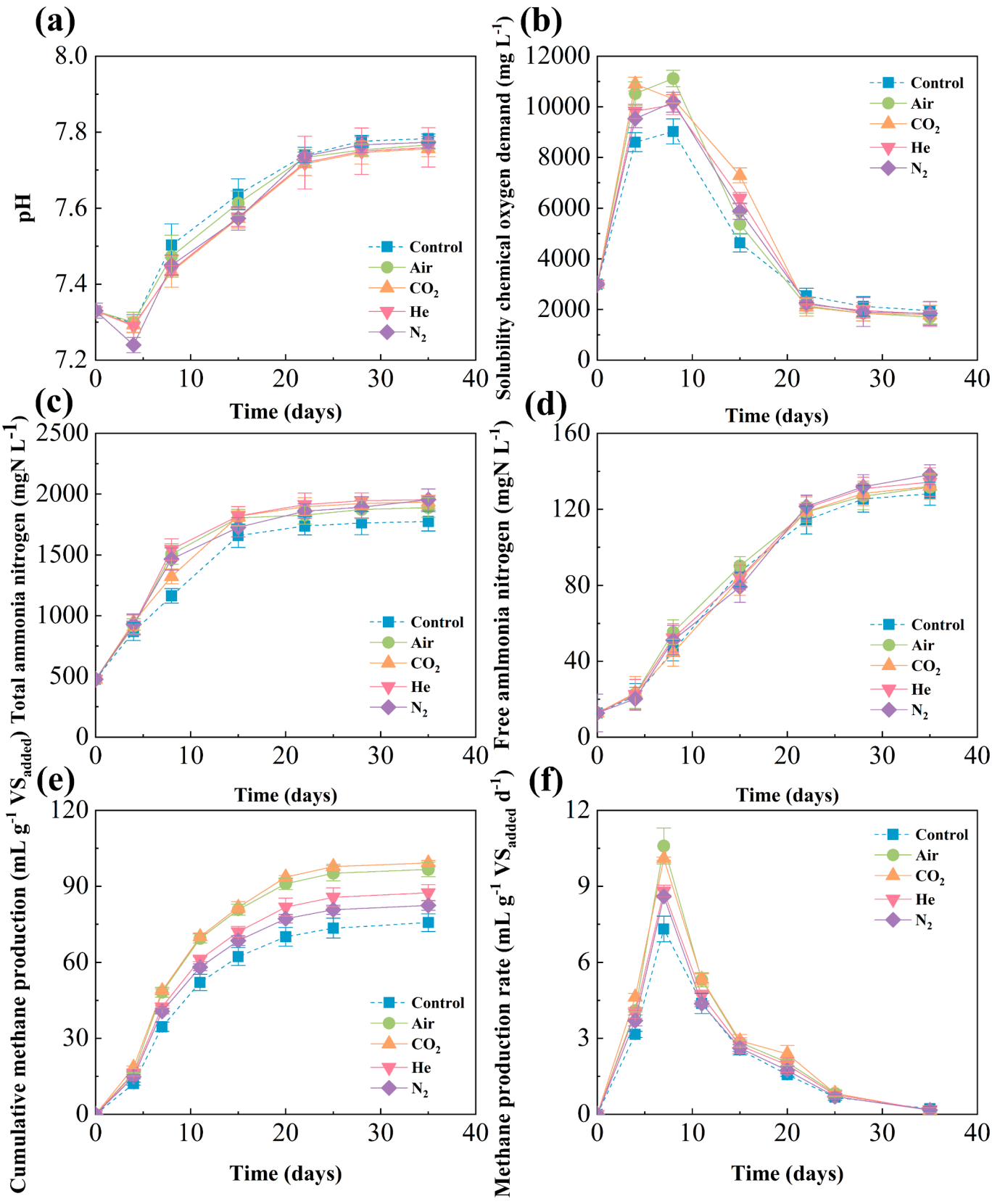


Fig. 2. Changes in pH (a), SCOD (b), TAN (c), FAN (d), cumulative methane production (e), and methane production rate (f) during anaerobic digestion.

(Fig. 2b); all of them showed a trend of increasing first, peaking between Day 4 and Day 8, and then decreasing; the same trend as the VFA changes (Wang et al., 2024). On Day 8, the SCOD of four digesters added with NBW (Air, CO_2, He, and N_2) were significantly increased by 23.1 % ($p < 0.01$), 14.2 % ($p < 0.01$), 11.7 % ($p < 0.01$), and 12.8 % ($p < 0.01$), respectively, as compared with control. In terms of the maximum SCOD, the optimal hydrolysis-promoting effect was found in the digester with Air-NBW. Air-NBW created a micro-oxygenic environment in the early stage of AD, which promotes the growth of facultative anaerobes and electron transfer between microorganisms, leading to the promoted hydrolysis of the substrate (Hou et al., 2022). The promoting effect of He-NBW and N_2-NBW showed less prominent, probably because the inert gases had less effect on the enzymes (Wang et al., 2023).

Ammonia is a product of protein hydrolysis that increases gradually with AD (Liu et al., 2022). The trend of TAN changes in all digesters was consistent, with an increase followed by a gradual leveling off, which was maintained at 500–2000 mgN L^{-1} (Fig. 2c). It is worth noting that TAN higher than 1500 mgN L^{-1} may inhibit methanogens (Wang et al., 2024). On Day 8, the TAN of digesters with added NBW (Air, CO_2, He, and N_2) were significantly increased by 22.8 % ($p < 0.01$), 12.1 % ($p < 0.01$), 24.7 % ($p < 0.01$), and 20.7 % ($p < 0.01$) as compared with the control. The addition of NBW can affect the activity of microorganisms because the properties of nanobubbles affect the growth and reproduction of microorganisms and their physiological activities (He et al., 2022). On Day 35, the digesters with added NBW (Air, CO_2, He, and N_2) significantly increased TAN by 6.4 % ($p < 0.01$), 9.0 % ($p < 0.01$), 10.1 % ($p < 0.01$), and 10.1 % ($p < 0.01$). The TAN difference (between added NBW and control) decreased from 160.3 to 181.3 mgN L^{-1} (Day 8) to 114.0–179.3 mgN L^{-1} (Day 35). This indicates that the addition of NBW can enhance sludge hydrolysis during the early stage of AD (Fan et al., 2021b). In addition, it has been reported that the addition of NBW can increase TAN in AD by 200–400 mgN L^{-1} on Day 4 and Day 10 (Fan et al., 2021a), which enhanced the hydrolysis in the early AD stage.

FAN is a part of TAN, which is significant inhibitory to methanogens (Capson-Tojo et al., 2020). All digesters showed similar trends in FAN, with FAN ranging from 0 to 150 mgN L^{-1} (Fig. 2d). FAN would increase with the AD, and FAN would show a trend of increasing and becoming later. When the FAN > 250 mgN L^{-1}, it would inhibit methanogens (Zhang et al., 2024). Consequently, the restraining influence of FAN on methanogens is confined.

The digesters with added NBW (Air, CO_2, He, and N_2) significantly increased ultimate methane production by 27.9 % ($p < 0.01$), 32.0 % ($p < 0.01$), 16.0 % ($p < 0.01$), and 9.2 % ($p < 0.05$) compared with the control (Fig. 2e and Table 1). This is because nanobubbles can effectively improve mass transfer, increase the activity of microbial community in the digesters, and expedite the breakdown of organic matter and the consumption of VFA (Hou et al., 2022). The highest cumulative methane production from CO_2-NBW was confirmed by VFA and SCOD measurements, which demonstrated the beneficial effect of NBW addition in optimizing the AD process of the sludge (Fig. 1 and Fig. 2b). The lower cumulative methane production in the digesters of He and N_2 may be a result of the higher TAN (Fig. 2c), as TAN above 1500 mgN L^{-1} inhibits the activity of methanogens (Wang et al., 2024). As reported, the addition of Air-NBW and CO_2-NBW increased cumulative methane production in the mesophilic digester by 20.8 % and 19.1 %, respectively (Fan et al., 2020). Existing studies have concentrated on a weaker impact of NBW, this can be credited to the increased concentration of substrate organic material and the relatively lower volume of NBW employed per unit of VS in the latter.

The methane production rate peaked on Day 7, NBW (Air, CO_2, He, and N_2) significantly increased the methane production rate by 44.8 % ($p < 0.01$), 37.9 % ($p < 0.01$), 20.4 % ($p < 0.01$), and 17.6 % ($p < 0.05$) compared with the control (Fig. 2e). Air-NBW enhanced the hydrolysis process in AD, and the organic matter available for microorganisms increased, leading to the highest methane production rate in Air-NBW on Day 7 (Fan et al., 2020). The cumulative methane production of He-NBW and N_2-NBW was close, suggesting that these two nanobubbles mainly served to promote hydrolysis (Wang et al., 2023). Besides the promotion of hydrolysis, the micro-oxygenic environment created by air nanobubbles and the hydrogenotrophic methanogenesis enhanced by CO_2 nanobubbles may further improve the cumulative methane production (Marcelino et al., 2023; Chuenchart et al., 2021). The Gompertz model verified that the addition of NBW could shorten the lag phase of the AD system (3.4–19.7 %); this may be because the nanobubbles had high stability and could exist in the digestion system for a long time. This stability helps to reduce the lag phase in the AD process (He et al., 2022).

Table 1
Parameters of the modified Gompertz model.

	Ultimate methane production, UMP (mL g^{-1} VS_{added})	Maximum methane production rate, R_m (mL g^{-1} VS_{added} d^{-1})	Lag phase, λ(d)	R^2
Control	74.3 ± 3.5 ^{a}	6.1 ± 0.6 ^{a}	1.9 ± 0.2^{b}	0.996 ± 0.001
Air	95.0 ± 2.8^{c}	8.3 ± 0.3^{c}	1.7 ± 0.1 ^{a,b}	0.995 ± 0.001
CO_2	98.1 ± 0.8^{c}	8.0 ± 0.3^{c}	1.4 ± 0.1 ^{a}	0.995 ± 0.001
He	86.2 ± 3.8^{b}	7.0 ± 0.2^{b}	1.5 ± 0.2 ^{a}	0.995 ± 0.001
N_2	81.1 ± 1.9^{b}	6.7 ± 0.2^{b}	1.5 ± 0.1 ^{a,b}	0.996 ± 0.001

Note: a, b, and c are labeled according to Duncan's test, and a significant difference between the two groups is considered to be significant at the $p < 0.05$ level when the labeled symbols between the two groups do not have repeated alphabets.

3.2. Microbial community

A relationship exists between the types and roles of the anaerobic microbial population involved in biogas production (Bahadur et al., 2024; Wu et al., 2021). It could be affected by the variety of feedstock, temperature, pH, TAN, and additional factors (Balasundaram et al., 2023). Assessing microbial diversity aids in identifying the composition of the community that results in modifications of the methanogens (Wu et al., 2023). The microbial community was analyzed for operational taxonomic units (OTUs), Shannon, Chao1, and Simpson indexes (see Supplementary Material). The OTUs, Shannon and Chao1 indexes of bacteria showed that the AD system with the addition of Air-NBW had a higher number of species than in the control. This is because the introduction of Air-NBW into the AD process increases the activity and diversity of the microbial community and promotes symbiotic interactions between different microbiota (Nguyen and Khanal, 2018).

Among the top ten archaea in relative abundance (RA), *Methanosaeta* and *Methanospirillum* are acetotrophic methanogens, while *Methanosarcina* is a facultative methanogen, and the other seven species are hydrogenotrophic methanogens (Fig. 3a) (Akindolire et al., 2022; Nguyen and Khanal, 2018). On Day 8, the RA of hydrogenotrophic methanogens in the NBW groups (62.5–90.1 %) was lower than that in the control (99.7 %); on Day 35, the RA of hydrogenotrophic methanogens in the NBW groups (33.5–54.7 %) was higher than that in the control (21.6 %). Hydrogenotrophic methanogens mainly rely on H_2 as an electron donor to reduce CO_2 to methane during their metabolic processes (Akindolire et al., 2022). On Day 8, NBW promoted sludge hydrolysis, thus the SCOD in the NBW groups was higher than that of the control. On Day 35, the cumulative methane production in the NBW groups was higher than that of the control, resulting in a lower SCOD in the NBW groups than that of the control. The difference in dissolved organic matter may cause the difference in the RA of hydrogenotrophic methanogens (Bahadur et al., 2024).

A reduction in the RA of *Methanobacterium*, *Methanobrevibacter*, *Methanocorpusculum*, and *Methanomassiliicoccus*, which were hydrogenotrophic methanogens (Akindolire et al., 2022; Nguyen and Khanal,

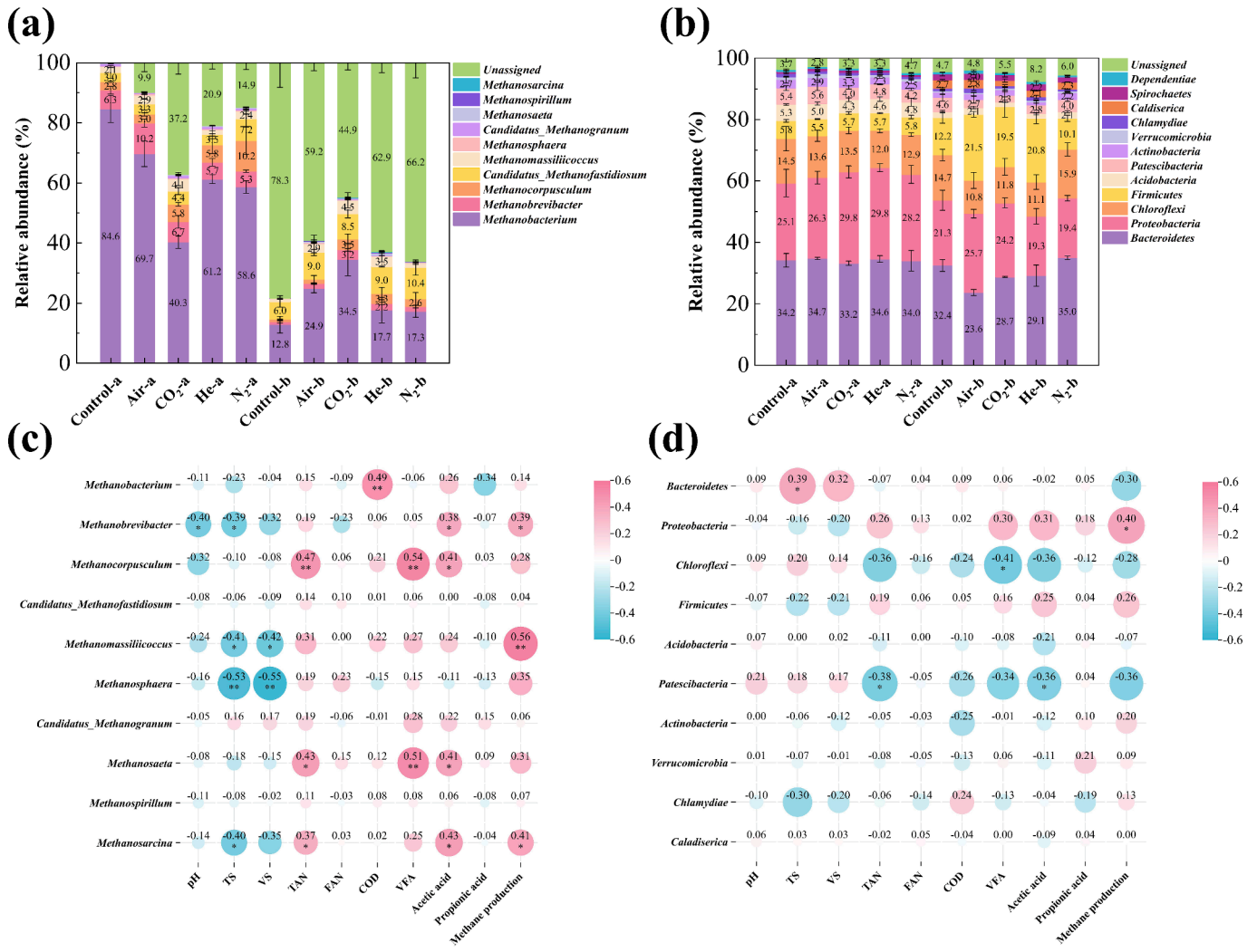


Fig. 3. Relative abundance of archaea (genus) and bacteria (phylum) during anaerobic digestion (a and b); Correlation analysis plots of environmental factors with archaea and bacteria respectively (c and d). Note: '-a' shows microorganisms on day 8 of anaerobic digestion; '-b' shows microorganisms on day 35 of anaerobic digestion. Correlation analysis was performed for each microorganism corresponding to the day's environmental factor.

2018), were observed in the comparison between Day 8 and Day 35 (Fig. 3a). *Methanobacterium* and *Methanocorpusculum* as methanogens have synergistic effects during AD and can engage with other methanogens and fermenters to facilitate the degradation and transformation of organic substances (Ziganshina and Ziganshin, 2023). After correlation analysis of archaea with anaerobic environmental factors, *Methanobacterium* demonstrated a substantial positive correlation with SCOD ($p < 0.01$) (Fig. 3c). *Methanocorpusculum* exhibited a marked positive correlation alongside TAN ($p < 0.01$) and VFA ($p < 0.01$). This suggests that *Methanobacterium* and *Methanocorpusculum* are more adapted to the VFA environment. *Methanobrevibacter* presented a substantial positive correlation with acetic acid content and cumulative methane production ($p < 0.05$). *Methanomassiliicoccus* exhibited a notable negative correlation in connection with TS ($p < 0.05$), and a substantial positive correlation among cumulative methane production ($p < 0.01$). This suggests that *Methanobrevibacter* and *Methanomassiliicoccus* play essential roles in the methanogenesis of AD. From the PCA plots of the archaea (see Supplementary Material), it can be seen that there are some differences in the archaeal community in both the hydrolysis stage and the methanogenic stage; this may be due to the different effects of different NBW on the microbial activity (Wang et al., 2023).

Compared with Day 8, a decrease in the relative abundance (RA) of *Bacteroidetes* was observed on Day 35. Based on the RA results compared on Day 8, N_2 (35.0 %) > control (32.4 %) > He (29.2 %) > CO_2 (28.7 %) > Air (23.7 %) (Fig. 3b). While the RA of *Bacteroidetes* was the least in Air, probably because Air-NBW was the best for AD and hydrolysis, so the hydrolysis stage ended the earliest in Air. This conclusion is in line with other findings in the literature that *Bacteroidetes* are protein-degrading bacteria (Meng et al., 2023). The RA of *Proteobacteria* decreased from 25.1-29.8 % to 19.3–24.2 % mainly due to the hydrolysis rate change. *Proteobacteria* encompasses bacteria that participate in the preliminary breakdown of biomass, chiefly producing propionate, butyrate, and acetate, which follow the findings presented in this paper (Ariesyady et al., 2007). Following comparative analysis, the RA of *Firmicutes* rose (from 5.5-5.8 % to 10.1–21.5 %). *Firmicutes* are anabolic nutrient bacteria acting on VFA to produce H_2 to be utilized by hydrogenotrophic methanogens, which is in line with the phenomenon (Park et al., 2023). *Bacteoidetes*, *Firmicutes*, and *Proteobacteria* can all be involved in the hydrolysis of lignocellulose (Meng et al., 2023). After correlation analysis of bacteria with anaerobic environmental factors, a notable positive correlation was observed ($p < 0.05$) between *Proteobacteria* and cumulative methane production (Fig. 3d); probably because *Proteobacteria* actively participated in the hydrolysis of AD (Ariesyady et al., 2007). The PCA of the bacteria showed some differences in the bacterial community during the hydrolysis stage, but the differences were not significant (see Supplementary Material); probably because this study was a mesophilic AD and the bacterial community changed more slowly (Siqueira et al., 2022). The bacterial community in the

methanogenic stage varied greatly and mainly were hydrolyzing bacteria, proving that NBW can indeed promote microbial activity (Kurz et al., 2021).

3.3. Dewaterability

The particle size distribution of digestate is intimately connected to its dewaterability (Zhang et al., 2024). The digesters with added NBW (Air, CO_2, He, and N_2) decreased digestate size (d50) by 30.8 %, 29.7 %, 31.2 %, and 23.0 % compared with the control (Fig. 4a). The addition of

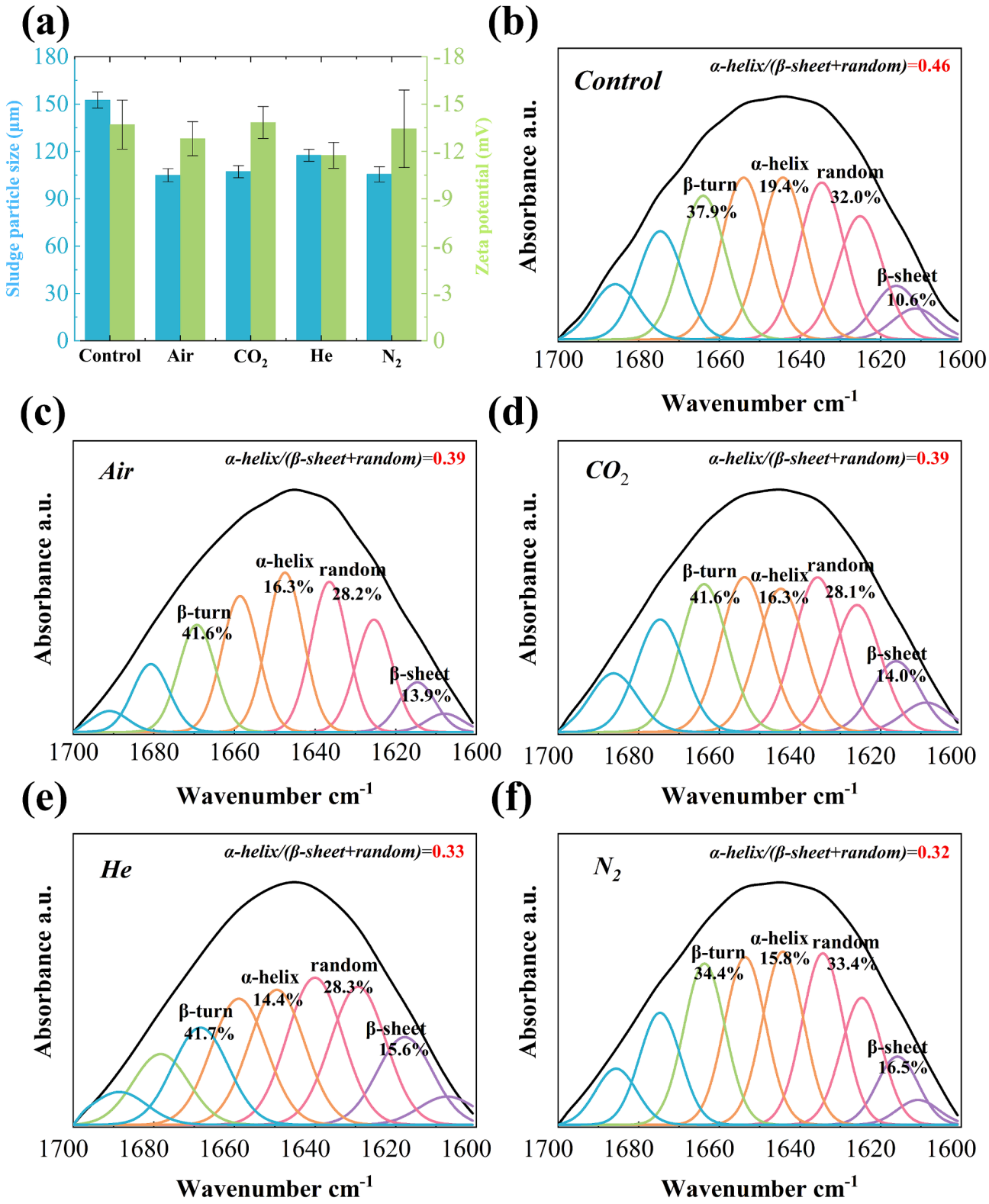


Fig. 4. Digestate particle size and zeta potential (a) and protein secondary structure analysis maps (b: control; c: Air; d: CO_2; e: He; f: N_2).

NBW enhances sludge degradation, which leads to a reduction in digestate particle size (He et al., 2022). It has been reported that the more thoroughly AD is carried out, the smaller the digestate particle size (Liang et al., 2022). When the digestate particle size exceeds 100 μm, the reduced dimensions of digestate particles contribute to enhanced dewaterability (Ward et al., 2023). Oversized digestate particles may have more water encapsulated inside, which is weakly bound to the particles and can be easily released during the dewatering process (Lin et al., 2019).

Zeta potential is a critical indicator for assessing the charge repellency of the digestate interface (Cao et al., 2023). At the end of AD, the zeta potential of control became −13.7 ± 1.6 mV, and that of Air, CO_2, He, and N_2 became −12.8 ± 1.1 mV, −13.8 ± 1.1 mV, −11.8 ± 0.8 mV, and −13.4 ± 2.5 mV (Fig. 4a). The addition of NBW increased the zeta potential of digestate, probably due to the high surface activity of the bubbles in the NBW, this modification affected the electrostatic distribution across the particle interfaces, resulting in elevated zeta potential (Kurz et al., 2021); while the addition of CO_2-NBW slightly increased the zeta potential of DS, probably because the tiny bubbles in the CO_2-NBW increased the collision frequency between the digestate particles, which led to a decrease in the zeta potential of DWS (He et al., 2022). The digestate zeta potential decreased from −20.6 mV to −29.4 mV due to pretreatment, leading to the deterioration of digestate dewaterability, a pattern similar to that of the present study (Cao et al., 2023).

There is a strong correlation between the hydrophobicity of digestate and protein secondary structure (Liang et al., 2022). The addition of NBW led to a general rise in the proportion of β-sheet and β-turn structures, with a marked reduction in the proportion of α-helix and random coil conformations (see Supplementary Material). In addition, the α-helix/(β-sheet + random) ratio is often used to characterize the compositional features of protein secondary structure (Zhang et al., 2024). A decrease in this ratio indicates a looser structure of the protein molecule (Estevam et al., 2024). The α-helix/(β-sheet + random) ratio of digestate in control was 0.46, while the ratios of the experimental group with the addition of NBW were 0.39, 0.39, 0.33, and 0.32, respectively (see Supplementary Material). The EPS structure of digestate gradually changed from “tightly ordered” to “loosely disordered”, and the loose structure of the protein in EPS exposed more internal hydrophobic groups (Hou et al., 2015).

EPS is a class of polymers secreted by microbial cells that are tightly bound to the cells (Zhang et al., 2024). EPS in digestate can be categorized into soluble EPS (S-EPS), loose layer EPS (LB-EPS), and tightly bound EPS (TB-EPS) (Hou et al., 2015). The trend of EPS within all digesters remained consistent, with a gradual decrease in EPS with AD (see Supplementary Material). As the AD progressed, the protein content in the S-EPS decreased by 61.1–65.8 % compared with that before AD, while the polysaccharide content increased and then decreased. The EPS in this study showed a decreasing trend, which could be attributed to the consumption of proteins and polysaccharides in the sludge by AD (Chuenchart et al., 2021).

The 3D-EEM plots of EPS extracted from digestate were investigated; the higher the fluorescence intensity of S-EPS, the worse the digestate dewaterability, and the higher the fluorescence intensity of TB-EPS, the better the digestate dewaterability (Simoni et al., 2024). Compared with the control, there was a reduction in complexine, tryptophan, fulvic acid, and soluble microbial byproducts in the digesters with the addition of NBW in the S-EPS (see Supplementary Material). The fluorescence region of control for TB-EPS was smaller than that of Air, CO_2, He, and N_2. In terms of the comparison of peak heights of humic acid, Air (329.3) > CO_2 (220.2) > He (158.0) > N_2 (154.9) > control (151.9). When more biopolymers from TB-EPS were extracted into S-EPS, the dewaterability was poorer (Chuenchart et al., 2021).

CST is an indicator for assessing digestate dewaterability (Zhang et al., 2024). On Day 35, the digesters with added NBW (Air, CO_2, He, and N_2) decreased CST by 18.3 % ($p < 0.01$), 16.5 % ($p < 0.01$), 15.1 % ($p < 0.01$), and 15.1 % ($p < 0.01$) compared with the control (Fig. 5a); indicating that the addition of NBW had a good improvement on the dewaterability of digestate. This may be due to NBW destroying the hydrophobic layer on the surface of digestate particles, thus favoring digestate dewaterability (Patel et al., 2021). Four proteins in S-EPS exhibited a favorable association with CST ($p < 0.05$), indicating that the dewaterability gradually deteriorated with increased protein content in S-EPS. Proteins are very attractive to water, and as the percentage of proteins in the EPS increases, the worse the dewaterability of the EPS (Chuenchart et al., 2021). The α-helix also demonstrated a constructive correlation with CST ($p < 0.001$), whereas β-sheet ($p < 0.05$) showed a negative correlation (Fig. 5e). The decrease in the ratio of α-helix/(β-sheet + random) indicates that the PN molecular structure is looser and enhances the hydrophobicity of digestate (Liang et al., 2022). Also, Air-NBW has the highest ratio in α-helix/(β-sheet + random), which indicates the best dewaterability of the digester with Air-NBW. The best improvement effect on DWS of AD with the addition of Air-NBW may be because Air-NBW can promote the degradation of organic matter and improve dewaterability (Fan et al., 2021b).

SRF serves as a critical measure for assessing the filterability of digestate (Ward et al., 2023). When the SRF of digestate is greater than 5×10^{12} m kg^{-1}, the digestate has low dewaterability (Zhang et al., 2024). The SRF of the initial sludge was $(6.5 \pm 1.0) \times 10^{12}$ m kg^{-1}, and after AD, the SRF of control increased to $(20.6 \pm 0.5) \times 10^{12}$ m kg^{-1}, and the dewaterability of digestate deteriorated. On Day 35, the digesters with added NBW (Air, CO_2, He, and N_2) decreased SRF by 29.5 % ($p < 0.05$), 33.0 % ($p < 0.05$), 32.8 % ($p < 0.05$), and 25.1 % ($p < 0.05$) compared with the control (Fig. 5c). Digestate particle size corroborates each other: the smaller the digestate particle size, the better the dewaterability (Fig. 4a). As the digestate particles become smaller, their specific surface area increases, this facilitates the expulsion of water and the separation from the digestate particles (Ward et al., 2023). SRF exhibited a marked positive correlation with two proteins in S-EPS ($p < 0.05$), α-helix ($p < 0.001$), β-sheet ($p < 0.05$), and random ($p < 0.05$) (Fig. 5e). It may be because the micro-nanobubbles in the NBW can take the digestate particles up in the process of rising to form the suspension state, which helps to the dispersion of the digestate particles and reduces their SRF (He et al., 2022).

MC represents the proportion of the weight of water within the digestate relative to the aggregate weight of the digestate (Zhang et al., 2024). On Day 35, the digesters with added NBW (Air, CO_2, He, and N_2) decreased MC by 19.5 % ($p < 0.05$), 16.7 % ($p < 0.05$), 14.6 % ($p < 0.05$), and 13.6% ($p < 0.05$) compared with control (Fig. 5d). Three proteins in TB-EPS exhibited a substantial inverse relationship to MC ($p < 0.05$), dewaterability was significantly reduced with the decrease of proteins in TB-EPS (Fig. 5e). Proteins in TB-EPS are involved in many biofilm functions, and a decrease in proteins may reduce the functionality of biofilms, thereby affecting water transport and dewaterability processes (Chuenchart et al., 2021). NBW presented a significant decrease in the MC of digestate, which might be related to the accelerated decomposition of complex organic matter and small molecules by the addition of NBW (Zinjenab et al., 2023).

Utilizing the collaborative validation from CST, SRF, and MC, it is determined that the dewaterability of digestate is compromised. In the dewaterability test after AD with NBW, CST, SRF, and MC were decreased by 15.1–18.3 %, 25.1–33.0 %, and 13.6–9.5 %, respectively. The addition of NBW reduced the digestate particle size by 23.0–30.8 %, and when the digestate particle size exceeded 100 μm. The reduction of digestate particle size helped to improve the dewaterability of the digestate (Ward et al., 2023). The introduction of NBW leads to an increase in zeta potential in the AD system, which usually means an increase in repulsive forces between digestate particles (Marcelino et al., 2023). This may result in the digestate flocs becoming more loosely packed, thus improving the dewaterability of the digestate. The addition of NBW reduced the α-helix/(β-sheet + random) ratio by 15.2–30.4 %, which loosened the protein secondary structure, exposed more internal hydrophobic groups, and improved the dewaterability of the digestate

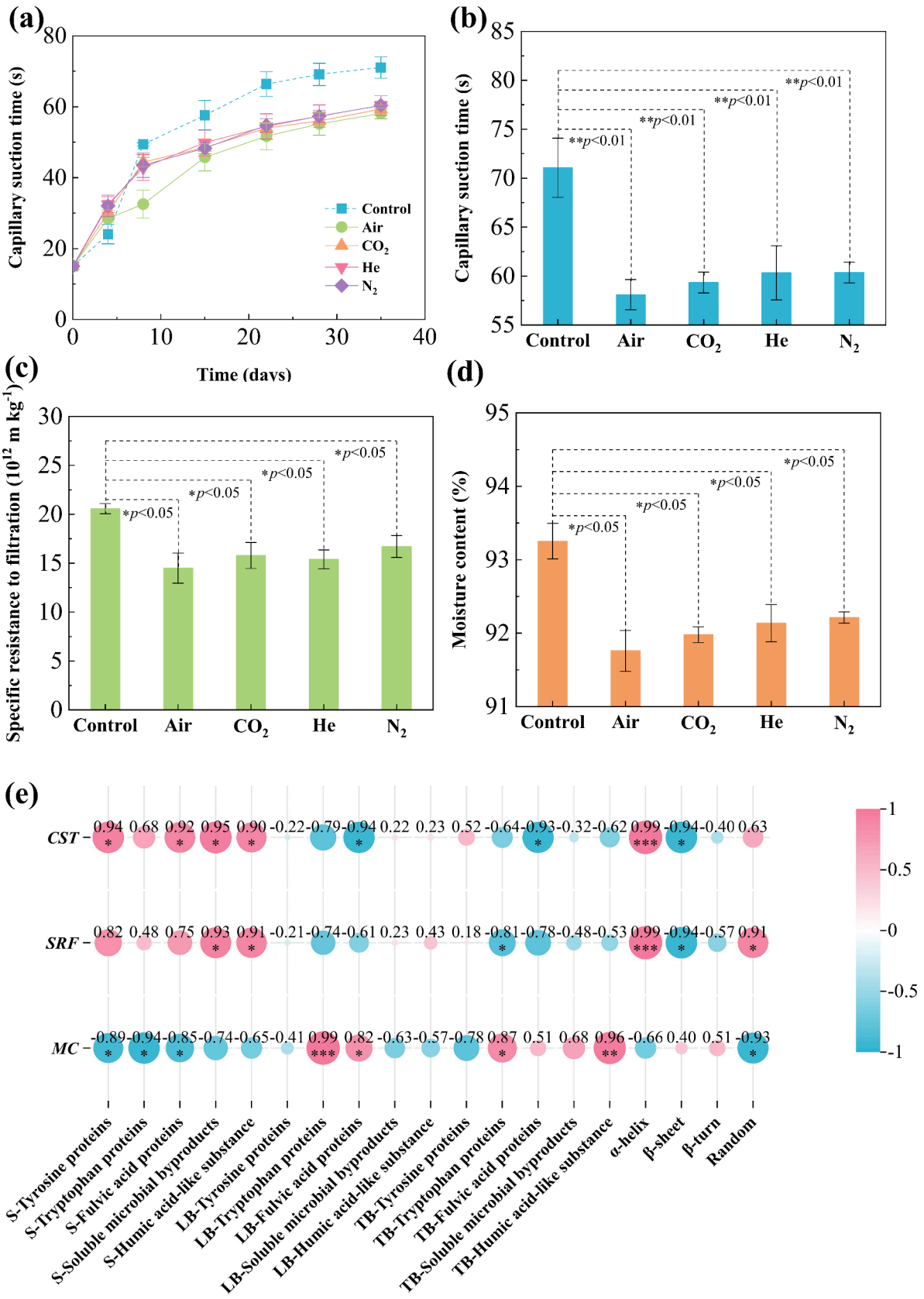


Fig. 5. Changes in CST during anaerobic digestion (a); CST (b), SRF (c), MC (d) at the end of anaerobic digestion; correlation heatmap between CST, SRF, MC, and 3D-EEM, the secondary structure of proteins (e).

(Estevam et al., 2024). The addition of NBW is an economically viable method to enhance the effectiveness of digestate compared with other treatment methods. This result is attributed to the unique properties of nanobubbles, such as high specific surface area and surfactant effect, which are conducive to enhancing the efficiency of AD in sludge (He et al., 2022). Digestate spiked with NBW can still be safely used for land improvement, thus realizing the resourceful use of DWS (Wang et al., 2023).

4. Conclusion

A positive influence was observed in cumulative methane production from DWS and the dewaterability of digestate with the addition of NBW (Air, CO_2, He, and N_2), in which the most prominent effect was found in Air-NBW. The addition of Air-NBW increased protein decomposition and the cumulative methane production was improved by 27.8 %. Simultaneously, the indicators of dewaterability, i.e. CST, SRF, and MC were decreased by 18.2 %, 29.6 %, and 19.5 % respectively, after 35 days of digestion, as compared with the control. Overall, Air-NBW is the most cost-effective additive for AD of DWS.

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Tianfeng Wang: Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Jie Wang:** Writing – review & editing, Writing – original draft, Resources, Investigation, Formal analysis, Data curation. **Jiazi Niu:** Visualization, Resources, Methodology, Investigation. **Peilin Guo:** Validation, Investigation. **Cheng Peng:** Investigation. **Rixing He:** Investigation. **Zhou Hui:** Investigation. **Wenqi Gao:** Investigation. **Qingfang Zhang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.130791.

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